



Habib University

Course Title: Digital Logic and Design (EE-172 / CS-130 / CE-222)

Instructors: Muhammad Farhan, Maria Waqas, Areeba Aziz Rajput

Assignment No: 2

Section:

Student Name:

Student ID:

Release Date: September 18th, 2025

Total points: 100

Due by: September 28th, 2025 11:59 PM

Points obtained:

Purpose:

The purpose of this assignment is to help you understand and apply the concepts and theorems of Boolean algebra for the implementation and simplification of logic functions. In addition, this assignment will also aid you to practice the tools and techniques used to minimize logic functions.

Instructions:

This assignment should be done individually.

All questions should be answered in **black ink only**.

Scan your answer sheet and upload it on LMS before the due date.

Grading Criteria:

Your assignments will be checked by instructor/TA.

You can also be asked to give a viva where you will be judged whether you understood the question yourself or not. If you are unable to correctly answer the question you have attempted right, you may lose your marks. No excuses/justifications will be entertained.

Zero will be given if the assignment is found to be plagiarized.

Untidy work or improper submissions will result in a reduction of your points.

Late submission penalty:

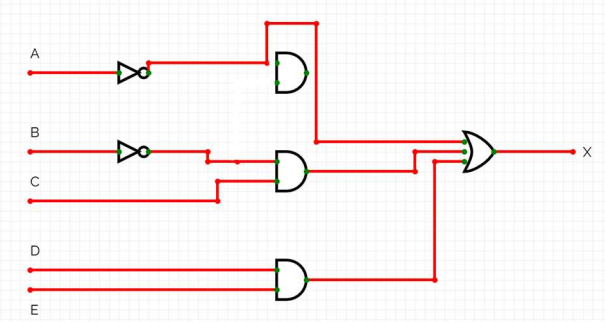
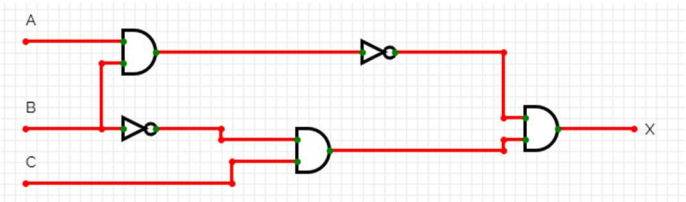
1-day late submission – 20% deduction of the maximum allowable marks

2-days late submission – 40% deduction of the maximum allowable marks

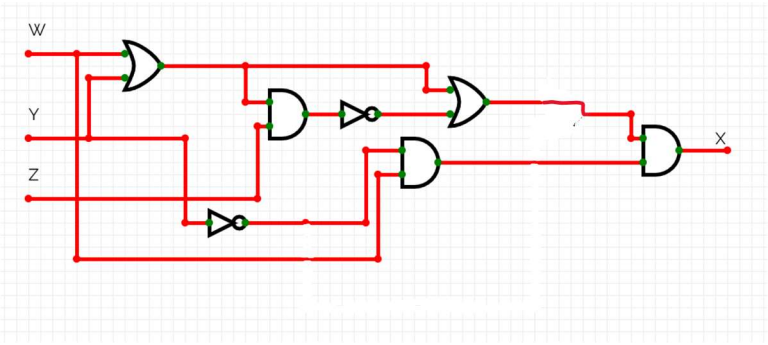
No submission will be accepted after two days of the original deadline

Q#	Questions	Pts
1	Faaz is trying to reach Gate 2 from Tapal during heavy rainfall. Each step he takes is stored as a 10-bit binary number. For him to not slip, each of his footsteps must be converted to Gray Code: - Design a logic circuit that converts a 8-bit binary number into Gray Code. - Simulate the logic on CircuitVerse (recommended) or logic.ly and add the screenshot that shows the conversion of 10110101 to its equivalent gray code. - Explain why he would have a chance of slipping if his footsteps aren't converted to Gray Code. (Hint: Look at consecutive Gray codes and Binary numbers)	2+2+1
2	Shehryar Rind is confused at the terrible drainage system of Karachi. He writes big Boolean equations to uncover why this is the case: - Find the complement of the below expression by De-Morgan's law. Verify your solution by truth table: $F = (x + y + z')(x' + y' + z)(w + x + y + z')(w' + x' + y + z)$ - Simplify the following expression using boolean algebra and mention the number of literals in the simplified expression: $(a + b')(c + d')(a' + c)(b + d)(a' + b' + c' + d')$ - Umar and Nigarish finish their DLD Quiz early. They then decide to compare their 12 quiz answers. Each T denotes a 1 and F denotes a 0: Umar's Answer key: TTFTFTTTFFTT Nigarish's Answer key: TFTTTTFFTF Convert each into a 12-bit Binary number and find the following: $A \otimes B$ $A \vee B$ A'	2+ 2+4x1
3	Professor Muhammad Farhan informed his students, that he will only pass them, if they fit his quality standards. The standards are defined by the expressions: $F1 = (a + b')(c + d)$ $F2 = (ab' + cd)(a' + d)$	2x2

	Prove that: a) The Boolean function $E = F1 + F2$ contains the sum of the minterms of F1 and F2. b) The Boolean function $G = F1.F2$ contains only the minterms that are common to F1 and F2.	
4	The EMTeam is busy preparing for the Freshers Beach Party, where a massive stage needs to be powered up for lights, sound, and projectors. Three switches are installed, each controlled by a different member. Wasiq, Erum and Hashim manage the lights, sound, and the projectors respectively. To avoid overloading the system and risking a total power failure, the stage can only be activated when exactly two of the three switches are turned ON at the same time. Any more or fewer than two will either leave the stage inactive or trigger a safety lockdown that may cause the event to shutdown. a) Create a truth table for the stage variable Z, which is 1 if the stage should be powered on. b) Write Minterms and Maxterms associated with the function and express the Boolean function for Z in both canonical forms. c) Simplify the function and redraw the reduced circuit.	3x2
5	The Professors of Habib University are unhappy with the collusion happening between students. To help prevent that, they are testing a new surveillance system installed in the classrooms. The alarm is based on 4 flags, a, b, c and d. Find the complement using De Morgan's Law of the alarm flag expression: $F' = (ab + cd')' + (a'b'c + d)' + (a + b' + c')' + (ab' + cd)' + (b + c + d')'$	3
6	Mohib designs a coffee selection algorithm for Sync Café. Right before he finalizes it, he is left with no space. He then decides to find a way to create more space. The logic for the algorithm is: $F(A, B, C, D) = (A' + B)(C + D')(B' + C + D)$ a) Identify the form of following equation. b) Simplify the following function to minimum number of literals to help Mohib create more space.	1+2
7	Dr. Strange's evil counterpart Strange Supreme, is on a quest to find M.O.D.O.K to help retrieve the Location of Kang the Conqueror so he could help him turn back time and save Christine Palmer. Unbeknownst to him, M.O.D.O.K is stuck in the Quantum Realm serving as a pet to Kang. He decides to use the	4x2

	Time Stone to quickly reach Valkyrie, who had promised a favor to him 3000 years ago. Unfortunately, the Time Stone was eaten by Tung Tung Sahur, which made Strange very mad, and he decided to use an ultimate spell on him which would finally turn him into an actual log of wood. The spell was surprisingly caught by a radar set by Dr. Reed Richards, listening to foreign anomalous activity. The radar converted the spell into digital logic and printed a sheet on which the following expression was written: $F = xy'z + x'y'z + w'xy + wx'y + wxy$ (a) Obtain the truth table of F. (b) Draw the logic diagram, using the original expression. (c) Use Boolean algebra to simplify the function to a minimum number of literals. (d) Draw the logic diagram from the simplified expression and compare the total number of gates with the diagram of part (b).	
8	Pakistan's bowling lineup is converted into Digital logic circuits and sketched by Shadab Khan on a piece of paper. To help visualize them and see their flaws, Coach Wasim Akram suggests checking NAND and NOR logic implementations of the circuits. a) Sketch NAND logic implementation to see their Spin-bowling flaws:  The diagram shows a NAND logic implementation with five inputs labeled A, B, C, D, and E. Input A is connected to a NOT gate, and its output is connected to one input of a 2-input AND gate. Input B is connected to another NOT gate, and its output is connected to the other input of the same 2-input AND gate. The output of this AND gate is connected to one input of a 3-input AND gate. Input C is connected to the other input of this 3-input AND gate. Input D is connected to the third input of this 3-input AND gate. The output of this 3-input AND gate is connected to one input of a 2-input OR gate. Input E is connected to the other input of this 2-input OR gate. The final output of the circuit is labeled 'x'. b) Sketch NOR logic implementation to see the Fast-bowling flaws:  The diagram shows a NOR logic implementation with three inputs labeled A, B, and C. Input A is connected to one input of a 2-input AND gate. Input B is connected to the other input of this 2-input AND gate. The output of this AND gate is connected to a NOT gate. Input C is connected to one input of a 2-input AND gate. The output of this AND gate is connected to another NOT gate. The outputs of these two NOT gates are connected to the inputs of a 2-input AND gate. The final output of the circuit is labeled 'x'.	4+2

9	Habib University is upgrading its campus security by switching to biometric and facial scanners. The digital lock system is defined by Boolean expressions that need to be verified and simplified to reduce wiring costs. a) Express the following in POS and simplify: $F(A, B, C, D) = (AB + C'D')'$ b) Express the following in SOP and simplify: $F(W, X, Y, Z) = (W' + X)(Y' + WZ)(XZ' + Y)$ c) Simplify the following SOP expression: $F(P, Q, R, S) = (P' + Q)(RS')(PQ + R)$ d) For the given complement, Write it as a product of max terms: $F'(M, N, O, P) = M'N + N'O' + M'OP'$ e) Simplify the following and implement with a two-level NAND circuit: $F(A, B, C, D) = (A' + B' + C)(A' + B)(B + C)$	3x4+4
10	The Reminisce Clothing pop-up store has a smart lighting system controlled by motion sensors (W,X,Y,Z). Due to electricity shortage, the logic must be simplified to minimize power use. a) Simplify using K maps: $F(W, X, Y, Z) = \Pi(0,2,5,8,9,11,13)$ b) Simplify with K maps using don't care conditions: $F(A, B, C, D) = \Sigma(0,2,6,10,14,15)$ $d(A, B, C, D) = \Sigma(1,5,9) \text{ (don't care conditions)}$	3x2
11	You are Dr. Amelia Brand, a scientist aboard the Endurance spacecraft. While navigating through the wormhole near Saturn to reach habitable planets in the Gargantua system, the ship's AI, TARS, has flagged an issue with the cryosleep pod stabilization system. The cryopods are critical for surviving the relativistic time dilation near Miller's planet. The Boolean function that controls the pod's stabilization thrusters is unnecessarily complex and is draining precious power reserves. It processes four variables: A (gravitational anomaly), B (time dilation sensor), C (oxygen level), and D (thruster alignment). The sum-of-products expression for the stabilization thrusters is: $X(A, B, C, D) = \Sigma(1,5,6,11,12,13,14)$ $d(A, B, C, D) = \Sigma(4)$ $Y(A, B, C, D) = \Pi(0,3,5,7,9,11,12,15)$	2x5

	a) Utilize Karnaugh Maps (K-maps) to simplify the Boolean functions X and Y, enhancing their efficiency and reducing complexity. b) Draw a logic circuit that replicates the behavior of the original Boolean functions X and Y after simplification. You can use CircuitVerse (recommended) or logic.ly and attach a properly labelled screen shot. Your goal is to reduce power draw in cryopod stabilization while preserving behavior, so efficient Boolean function simplification and circuit design are crucial for your research.	
12	You are hired as the scoreboard system designer for the very hyped futsal match between Hamza 11 and Salman 11. The electronic scoreboard system relies on a control logic function F(W, X, Y, Z) to register fouls, goals, and time-outs in real time. However, while production, you realize that there is a shortage of hardware modules in the sports complex and the original circuit is too bulky to build. To keep the match fair and the system responsive, you must simplify the circuit design so it can run smoothly on the limited setup and constraints.  Based on that constraint, you decided to reduce the number of gates in the system: - Use Boolean algebra techniques to get the simplified expression of the given circuit. - You are skeptical in your simplification and want to verify if the obtained expression is correct. For that, you make the truth table of the circuit above. - Using the truth table, simplify the circuit using K-maps. - Verify that the simplified expression obtained is the same as the one obtained in part a by drawing the simplified circuit. Is the number of gates used after simplification the same as the one used originally?	4x3
13	You are Master Yoda, hidden on Dagobah during the Galactic Civil War. There, you are trying to decipher a stolen Imperial	4+3x3

	holocron containing the logic for the second Death Star's shield generator bypass and hyperdrive disruptor. To help the Rebel Alliance's assault on Endor, you must simplify the Boolean functions to reduce power demands on an X-wing's onboard computer. You're given the following Boolean function that represents the logic behind the X-wing's control system: $F(W,X,Y,Z) = \sum(1,2,6,7,8,13,14,15)$ $d(W,X,Y,Z) = d(0,3,5,12) \text{ (don't care conditions)}$ - Using Karnaugh Map (K-map) reduction techniques, simplify the given Boolean function to minimize power consumption while maintaining the circuit's functionality. - Draw a circuit diagram for the simplified design. - Provide a breakdown of the number of terms and the number of literals in each term for both the original and reduced designs. - In your quest to reduce the power demands and help speed-up the assault, you've decided to replace all the various gate types in your circuit diagram from (b) with a combination of NOR gates, which are renowned for their insane energy efficient design. Create a NOR gate representation of your circuit design.	
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